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Advanced Multivariate Statistics Methods

A MULTILEVEL AND ROBUST INVESTIGATION OF E-COMMERCE DELIVERY LEAD TIMES

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TABLE OF CONTENT

- ▶ **01 Motivation & Research Questions**
- ▶ **02 Data Overview**
- ▶ **03 Methods**
- ▶ **04 Results**
- ▶ **05 Discussion**



MOTIVATION & RESEARCH QUESTIONS

Motivation & Research Questions

Delivery Lead Time Drives E-commerce Satisfaction, Yet Lacks Robust Statistical Analysis

1

Key service quality driver

Core dimension of E-commerce service quality and customer satisfaction

2

High variability

Large, unexplained variation in lead times across sellers and regions → need reliable ETAs

3

Research gap

Few studies use multilevel + robust + bootstrap to decompose and stress-test delivery lead times



Motivation & Research Questions

Research Questions

1 RQ1: Variance

How much variability in delivery lead time is attributable to differences between orders, sellers, and customer states?

2 RQ2: Fixed Effects

How do key geographic and shipment characteristics (routing, freight, value) influence delivery speeds?

3 RQ3: Robustness

To what extent are findings stable when accounting for outliers using robust estimators and bootstrap inference?

DATA OVERVIEW

A black and white photograph of a city street scene. In the foreground, a person is skateboarding away from the camera. They are wearing a dark tank top, shorts, and sneakers, and have a backpack. To their right, a person is riding a motorcycle, also away from the camera. The motorcycle has a license plate that reads "52-K9 4077". In the background, there are multi-story buildings with balconies and awnings. A large blue circle is overlaid on the image, centered on the skateboarder. The text "DATA OVERVIEW" is written in white, bold, sans-serif capital letters across the center of the circle.

Data Overview

Landscape of Olist Brazilian E-commerce

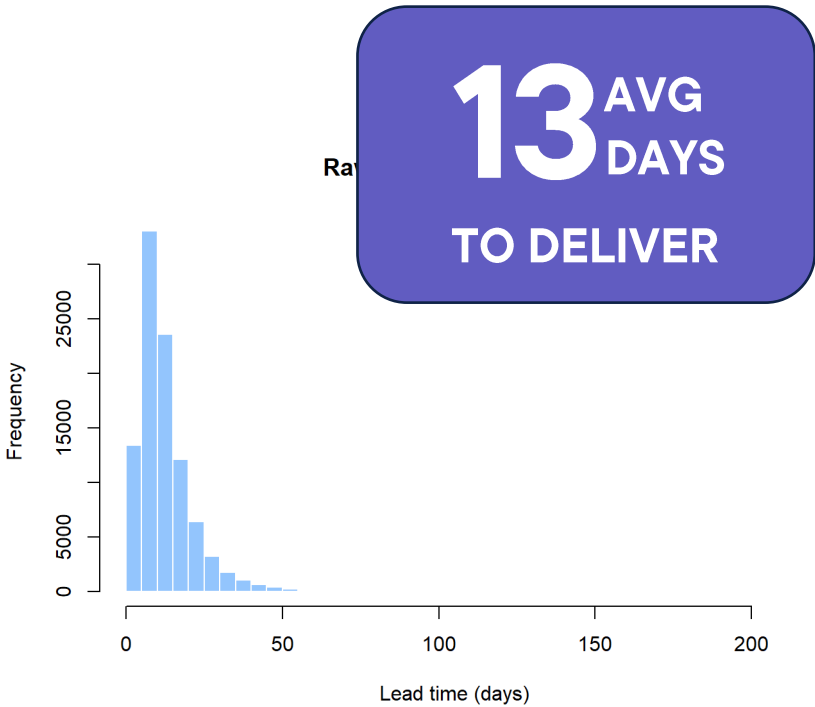
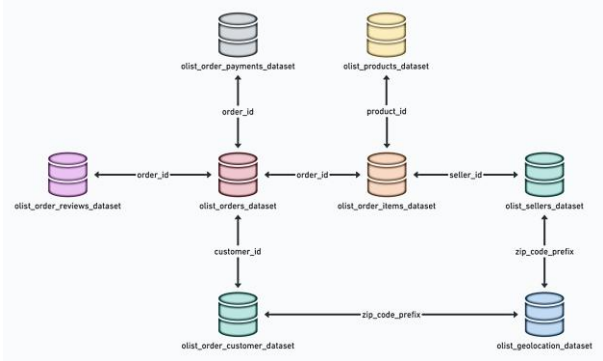
9
DATASET

27
STATES

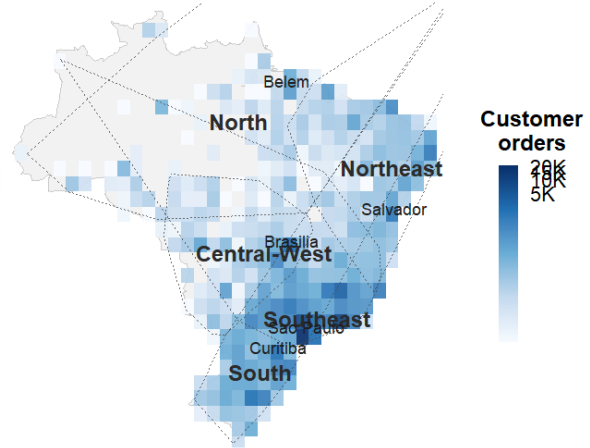
52
VARIABLES

3K
SELLERS

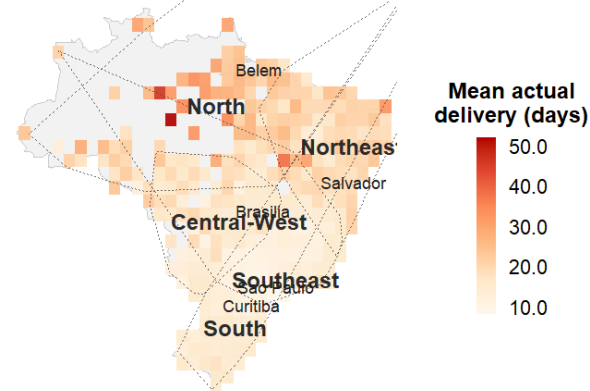
99K
ORDERS



Customer Order Density Heatmap



Mean Delivery Lead Time Heatmap



Data Overview

Variables

OUTCOME



DELIVERY LEAD TIME

order placement → delivery

$$y_{ij} = \log (1 + \text{lead_time_days})$$

KEY PREDICTORS



Geography

seller state, customer
state, same-state/same-
region indicators



Shipment patterns

weight, volume, freight
cost, items, order value

COVARIATE GROUPS



Time

Weekday
Season: DJF, MAM, JJA, SON



Product themes

Search vs Experience
Hedonic vs Utilitarian

METHODS

A black and white photograph of a city street scene. In the foreground, a person is skateboarding away from the camera. They are wearing a dark tank top, shorts, and sneakers, and have a backpack. To their right, a person is riding a motorcycle, also away from the camera. The motorcycle has a license plate that reads "52-K9 4077". In the background, there are multi-story buildings with balconies and awnings. A large blue circle is overlaid on the image, and the word "METHODS" is written in white capital letters across the center.

Three Analytical Challenges Require AMS Methods

Clustered Data

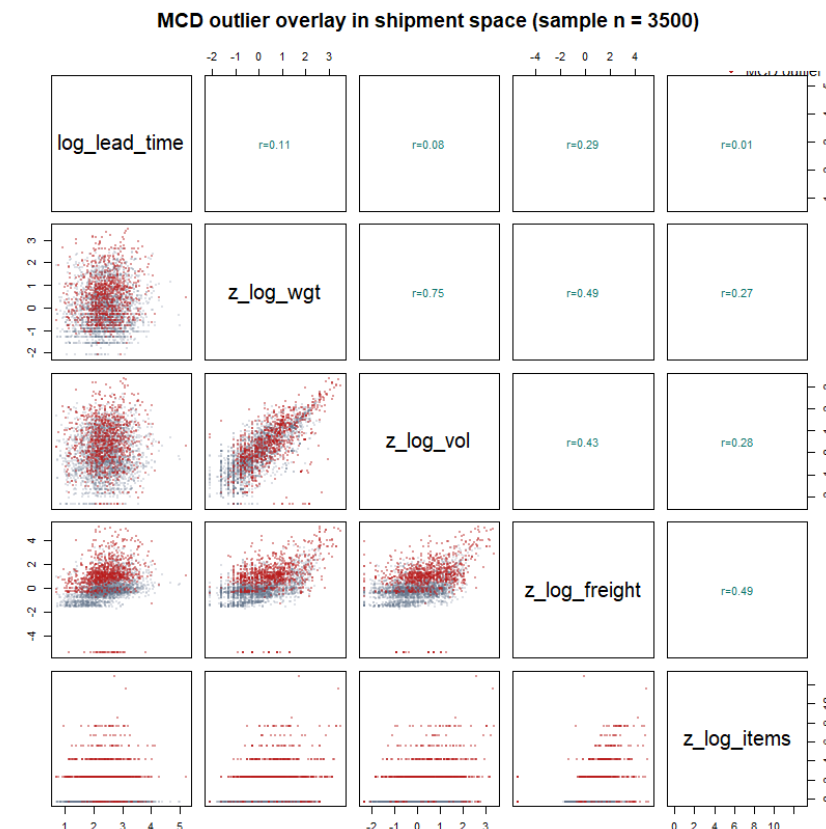
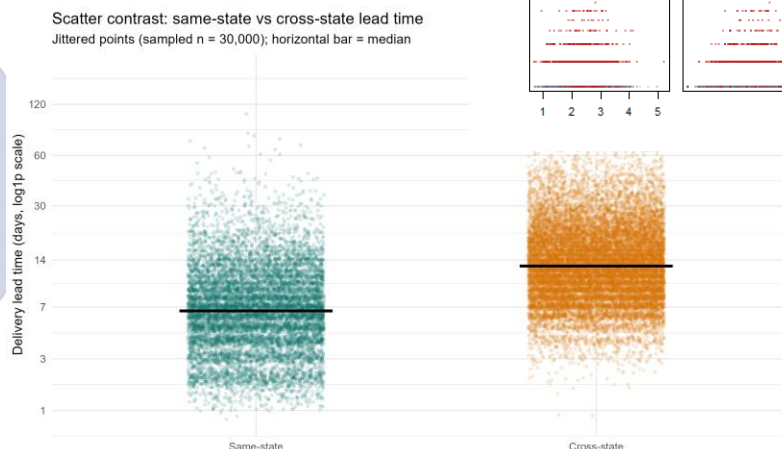
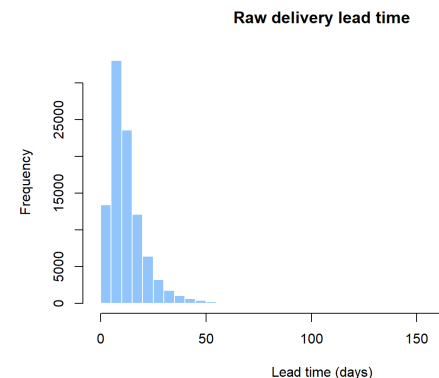
- Orders are clustered within sellers and customer states → residuals are correlated within clusters → OLS independence assumption is violated

Heavy right tail

- Long delays and multivariate shipment outliers (~9.6% flagged by MCD), a heavy upper tail reaching above 200 days

Latent heterogeneity

- Different shipment regimes (light vs heavy, cheap vs expensive) may show different effects



Integrated AMS Workflow



MLMM

Multilevel linear mixed models
with crossed random
intercepts for sellers and
customer states (MO–M3)



Model-based clustering (Gaussian mixtures)

on standardized log shipment
features to define five shipment
regimes



Robustness

Trimming, robust mixed-
effects estimation (M-
estimators), MCD diagnostics



Parametric Bootstrap

Validate Wald CIs under
unbalanced clusters and
heavy tails

Each method addresses a different failure mode;
joint use improves **CREDIBILITY**

Methods

Model Progression M0→M4 Tests Each Component

The baseline RE model M0:

$$y_{ijk} = \beta_0 + u_j + v_k + \varepsilon_{ijk}$$

y_{ijk} is log lead time for order i , seller j and customer state k

u_j : seller random intercept

v_k : customer state random intercept

M0

- Null: random intercepts only (seller + state)

M1

- + Routing (same-state, same-region)

M2

- + Shipment covariates (freight, items, value, weight, volume)

M3

- + Season, purchase year, product theme

M4

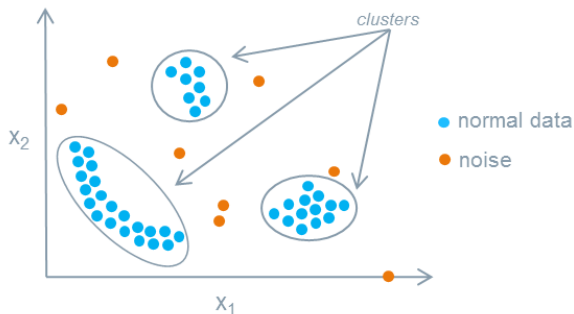
- + Cluster main effects & interactions (regime heterogeneity)

PROGRESSIVE LADDER shows how each block contributes to variance reduction

Methods

Model-Based Clustering for Regime Discovery

Why Cluster?



1

Heterogeneity Effect

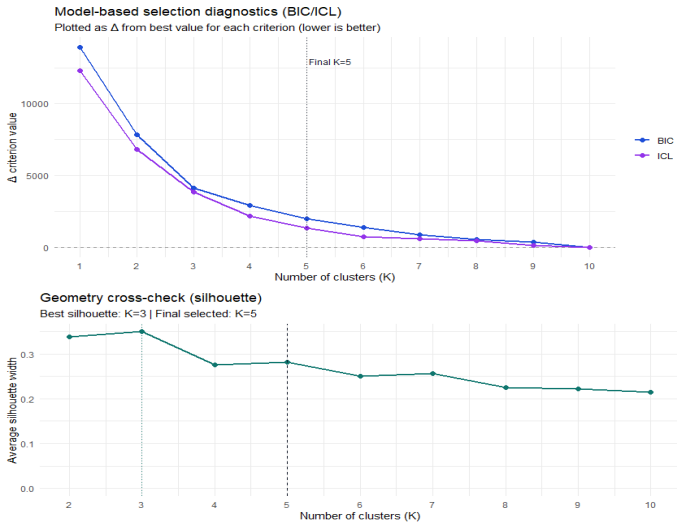
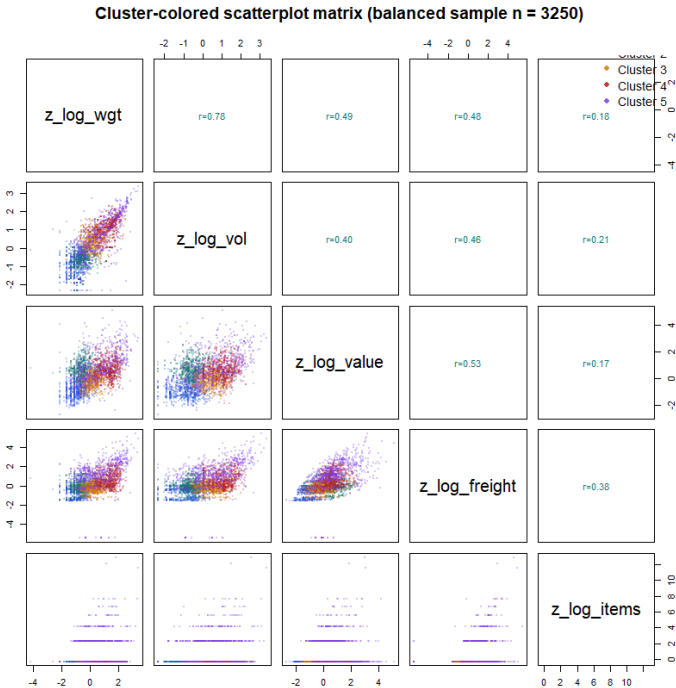
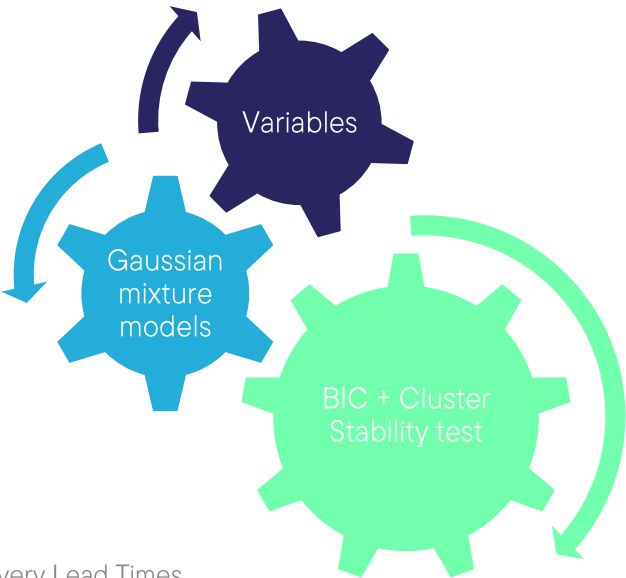
Suspect effect heterogeneity: routing/freigh effects may differ by shipment type

2

Reveal latent regime

Pre-cluster orders by shipment profile to reveal latent regimes

Clustering method



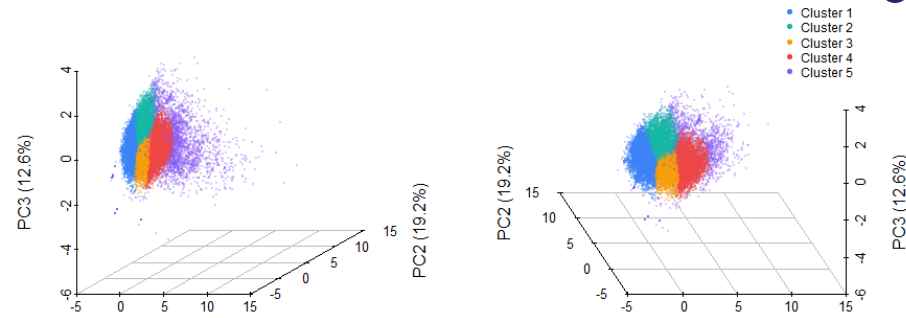
Methods

Five shipment regimes

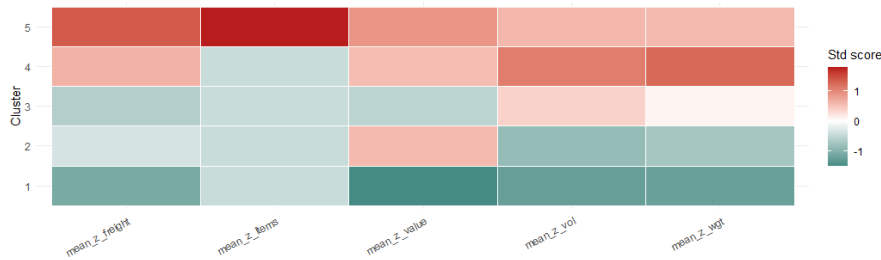
Pre-model clusters in PCA space (static 3D views, sampled n=25000)

View A

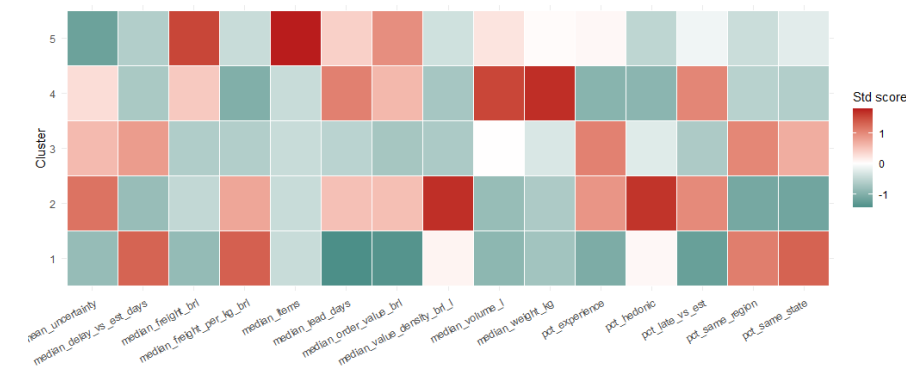
View B



Shipment profile heatmap
Mean z shipment features by cluster



Operational KPI profile heatmap
Interpretable KPI metrics standardized within feature



C5. High-freight, multi-item orders

high freight, high item count, and higher order value, with lower local-routing shares than C1/C3

C4. Bulky/heavy shipments

bulky/heavy profile (higher volume and weight), lower same-state share, and higher late-rate.

C1. Local, low-load basics

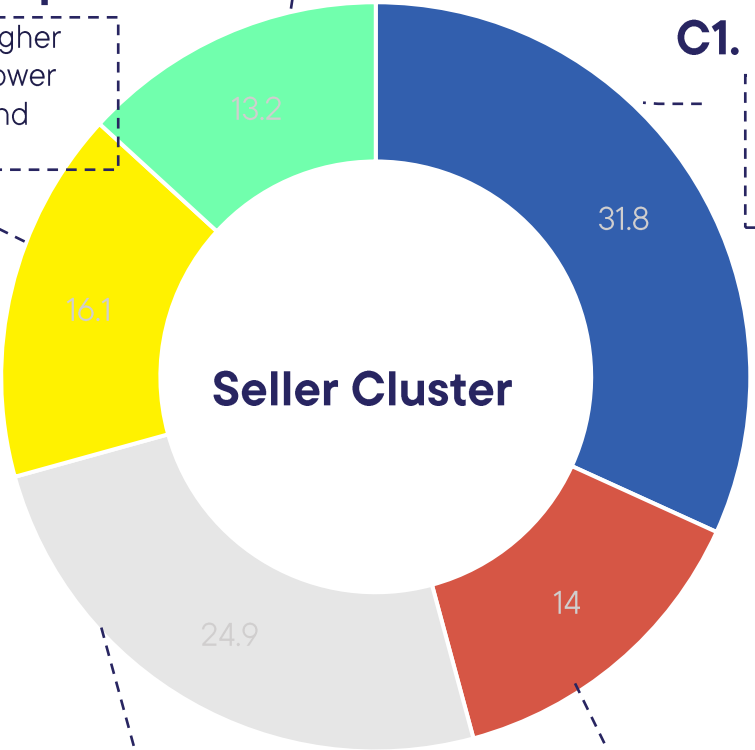
low-load/low-value orders, high same-state and same-region shares, and generally low uncertainty and late-rate

C3. Mid-load, balanced profile

mid-load and comparatively balanced profile, with moderate local-routing shares and intermediate delay behavior.

C2. Small but value-dense

physically small but value-dense, hedonic-leaning profile with lower local-routing shares and a higher late-rate



KEY FINDINGS

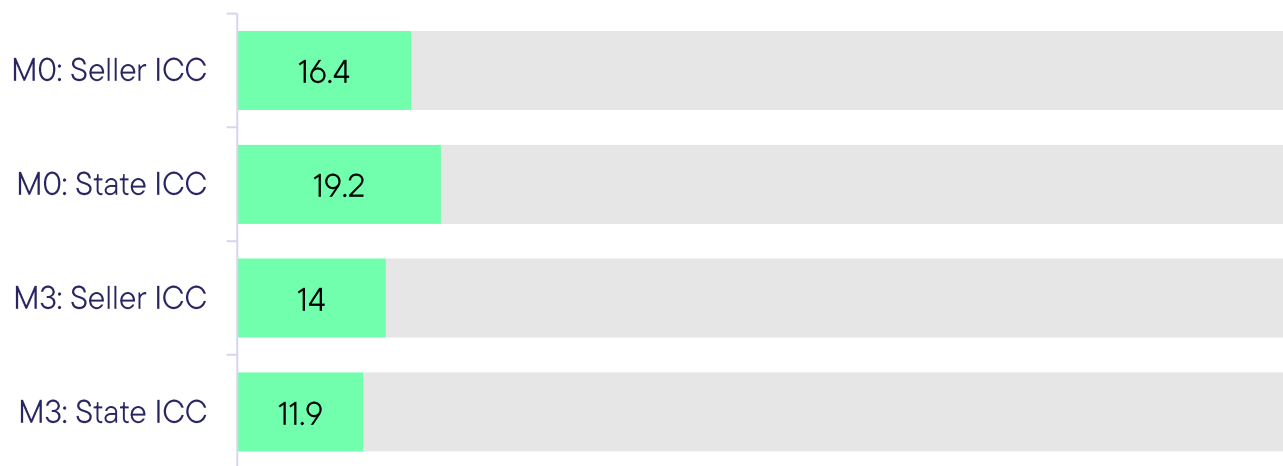


Results

RQ1: Where Does Variability Live?

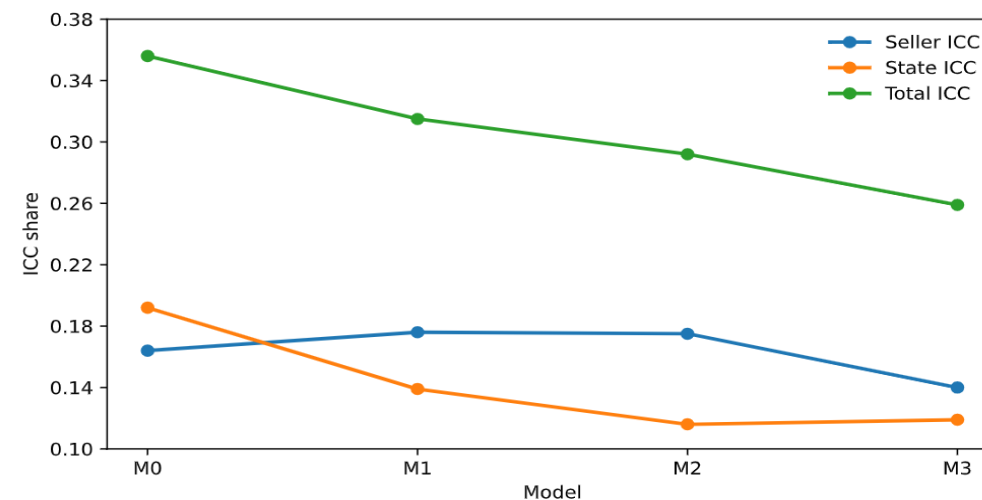
Seller and destination effects remain material → multilevel (crossed random effects) specification is still needed

Before vs After Adding Routing, Shipment, and Time Controls



Before controls (M0):

- **35.6% of variance** sits at seller/state levels (16.4% + 19.2%)
- State ICC slightly higher than seller ICC → geography matters as much as seller operations



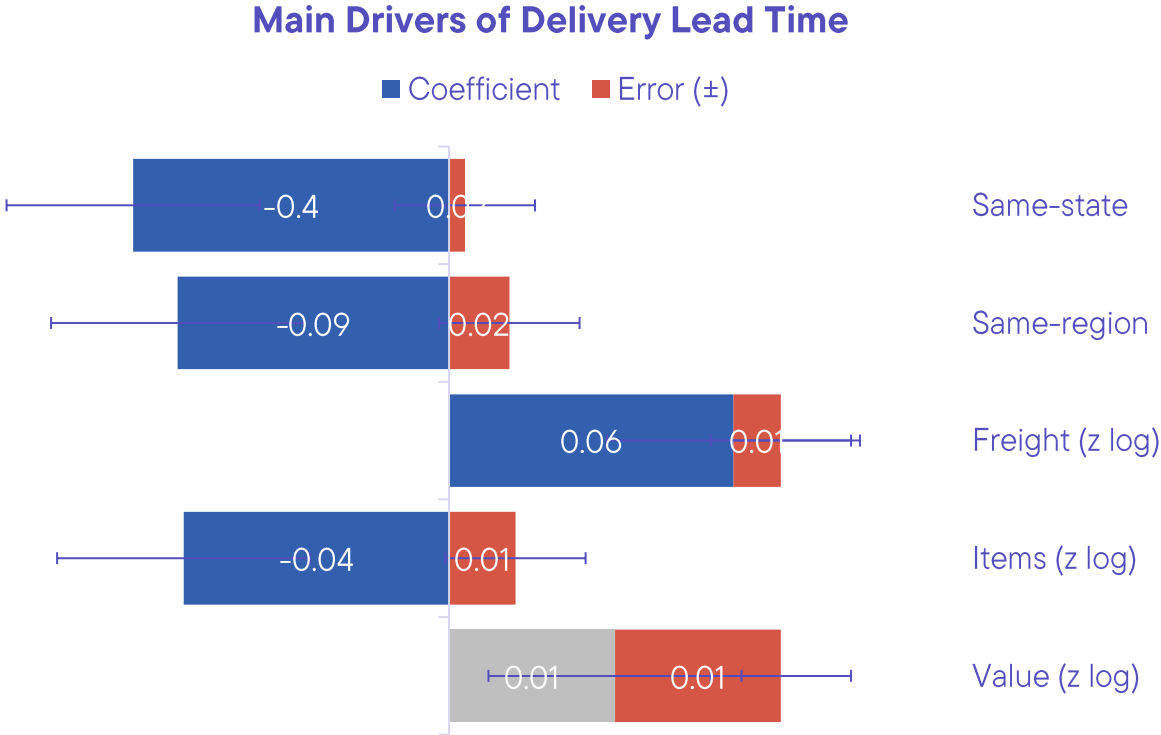
(b) Variance decomposition progression across M0–M3.

After full controls (M3):

- ICCs drop to **25.9% total** (14.0% + 11.9%)
- **Routing, shipment, and time** controls **explain ~10** percentage points of clustering
- But **seller & state heterogeneity remain non-negligible** → multilevel structure still justified

RQ2: What Drives Lead Time?

Same-state routing cuts delivery time by ~33%; freight cost slows it down



Key insights:

- Routing is the dominant operational lever
- Higher freight = longer lead time
- More items = slightly faster (batching/different channels)

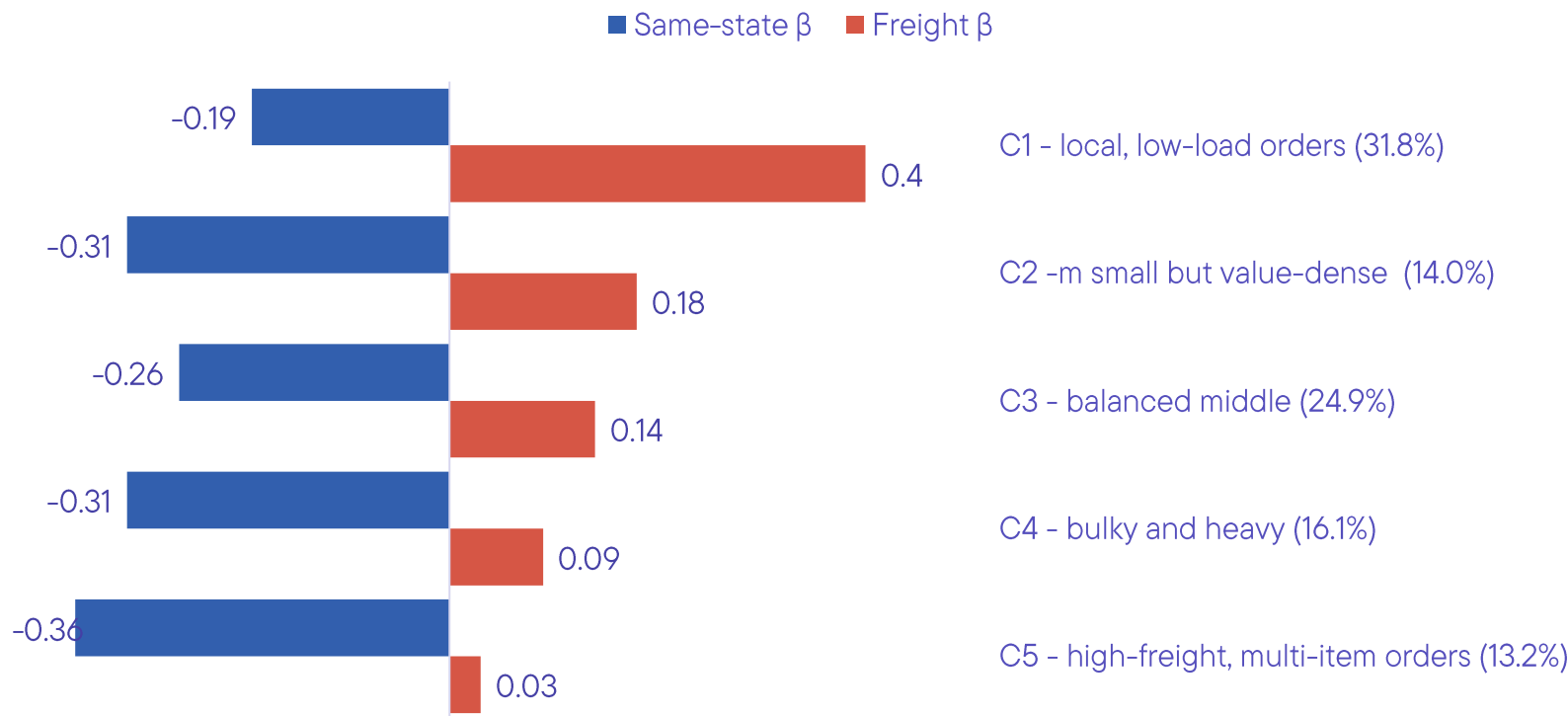
**Effect on log(lead time)*

Key Findings

RQ2: Effect Heterogeneity Across Shipment Regimes

Same effect direction everywhere, but magnitude varies 2× across regimes

Effect Heterogeneity Across Shipment Regimes



Key insights:

- Direction stable: same-state always negative, freight always positive
- Same-state effect ranges from **-0.19** to **-0.36**
- Freight sensitivity strongest in C1, weakest in C5
→ One-size-fits-all SLA misses regime-specific dynamics

Results

Model Comparison: Progressive Improvement

Routing gives the biggest jump; heterogeneity (M4) adds 2.4 pp in marginal R² and slashes AIC by 1,383

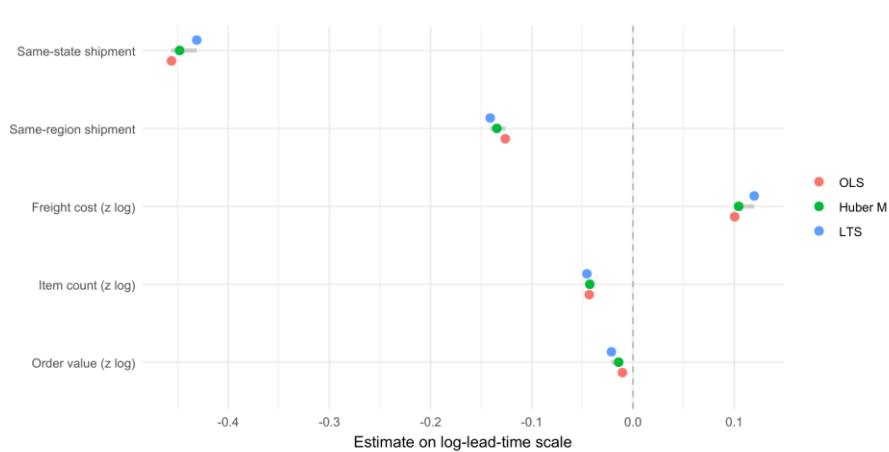
Model	Fixed Predictors	Marginal R ²	Conditional R ²	AIC	BIC	ΔAIC vs M3
M0	(None)	0.000	0.356	—	—	—
M1	+ Routing	0.139	0.382	—	—	—
M2	+ Shipment	0.160	0.397	—	—	—
M3	+ Time/Theme	0.223	0.424	117,189	117,369	<i>baseline</i>
M4	+ Cluster hetero	0.247	0.428	115,806	116,099	-1,383

- **Biggest gain:** M0 → M1 (routing alone explains 13.9% of variance)
- **Shipment profile refines:** M1 → M2 adds 2.1 pp
- **Controls matter:** M2 → M3 adds 6.3 pp (season, year, themes)
- **Heterogeneity improves fit decisively:** M4 vs M3 on common sample:
 - LRT: $\chi^2(8) = 1,348$, $p < 2e-16$
 - $\Delta AIC = -1,383$, $\Delta BIC = -1,270$
 - Marginal R² rises from 0.223 → 0.247
- **Conditional R² stays ~0.42–0.43:** Most variance still explained by hierarchy + fixed effects

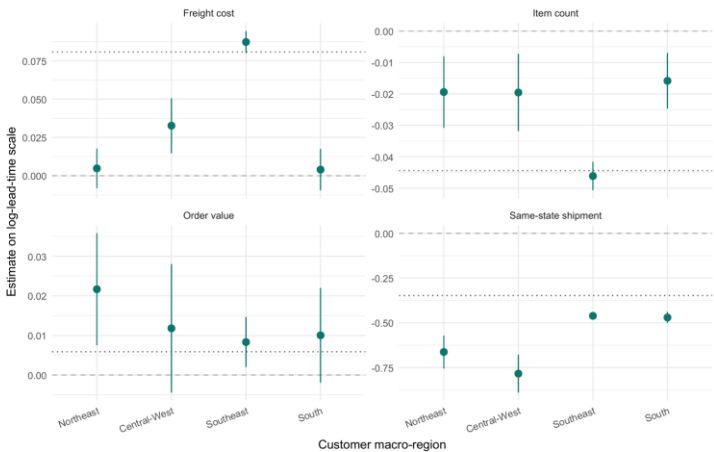
Results

RQ3: Are Findings Robust?

Same-state effect consistently negative across all macro-regions, not driven by one geography

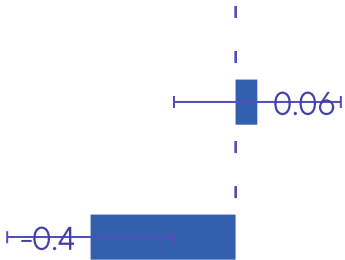


(a) Classical vs robust coefficient estimates.



(b) Regional mixed-model estimates vs national reference.

Bootstrap CI Wald CI Point



Robustness Check	Metric	Value	Threshold	Note
Robust estimators (MM, LTS)	Max Δ vs OLS	0.031	< 0.05	maximum absolute deviation from OLS is less than 0.05
Mixed-model variants	Max Δ vs M4	0.091	< 0.10	effect signs stay the same and magnitudes don't change by more than 10%
Regional submodels	Same-sign share	100%	> 90%	at least 90% of regions show the same sign** as the national estimate
Bootstrap vs Wald CIs	Overlap share	100%	> 80%	bootstrap percentile CIs overlap Wald CIs by at least 80%

- OLS (red) = Standard mixed model
- Huber M (green) = Robust to moderate outliers
- LTS (blue) = Highly robust (50% breakdown point)

DISCUSSION

A black and white photograph of a street scene in Vietnam. In the foreground, a person with long hair, wearing a dark tank top, shorts, and sneakers, is riding a skateboard. They are carrying a backpack and have a helmet on their head. To their right, a person wearing a plaid shirt and a helmet is riding a motorcycle. The motorcycle's license plate reads "52-K9 4077". In the background, there are multi-story buildings with balconies and awnings. A sign on one of the buildings reads "KIEN" and "CHINH HANH CONG". The entire image is framed by a large blue circle, and the word "DISCUSSION" is written in large white capital letters across the center.

Key Findings

GEOGRAPHY
(33% faster)

Same-state shipments are roughly one-third faster than cross-region shipments; same-region also helps but less.

SHIPMENT
(±5-8% effects)

Higher freight cost slows deliveries, while orders with more items are delivered slightly faster.

Regime-based clustering shows that effect size varies across shipment profiles, even when directions stay stable.

SELLER/STATE
(26% variance)

Even after controls, about 26% of variance remains between sellers and customer states ($ICC \approx 0.26$)

Discussion

Limitation & Future Direction

1

Single platform (Olist),
Brazil only

2

2016-2018 data
(pre-COVID logistics)

3

Missing: firstmile, lastmile
timestamp, logistic
operational metrics

Future Extension

Cross-
market
replication

Dynamic
models (time
effects)

Causal
Inference

Thank You.



APPENDIX

Appendix

Choosing the Number of Clusters (K)

K = 5 balances parsimony and operational interpretability

Fit Gaussian mixture models

- Test K = 3, 4, 5, 6, ..., 10 on standardized log shipment features

Evaluate BIC & ICL

- → Both improve up to boundary; no interior optimum

Compute silhouette (k-means)

- → Highest at K = 3 (≈ 0.35); K = 5 acceptable (≈ 0.28)

Check guardrail metrics

- → Posterior uncertainty, min cluster size, balance
- → K = 5 avoids tiny clusters, acceptable uncertainty

Assess stability (subsample ARI)

- → K = 5 shows highest stability (mean ARI ≈ 0.96)

Stability
Prioritize
Final decision

K = 5

